

Series XXV – Article XXV.3: Reduction to 3-SAT and Spectral Hamiltonian for Vizing’s Conjecture

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May 2026

Abstract

We complete the spectral reduction for Vizing’s conjecture. For fixed graphs G, H with at most N_0 vertices, we build a circuit $C_{G,H}$ that tests whether there exists a dominating set D of $G \square H$ with $|D| < \gamma(G)\gamma(H)$. We convert it to 3-SAT $\Phi_{G,H}$, build the spectral Hamiltonian $H_{G,H} = \hat{V}_{G,H} + \Delta$ on the hypercube, and verify the four pillars (P1–P4). The D-RSP algorithm will decide satisfiability in polynomial time.

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1 Introduction

Articles XXV.1 and XXV.2 provided the asymptotic bound and the domination circuit. Here we assemble the circuit for the Cartesian product, reduce to SAT, and build the spectral Hamiltonian.

2 Circuit for the Cartesian product

For fixed G, H , the vertex set of $G \square H$ has $n_G n_H$ vertices. A subset D is represented by $n_G n_H$ bits. The domination condition for each vertex (i, j) is:

$$\text{dom}_{i,j} = x_{i,j} \vee \bigvee_{i' \in N_G(i)} x_{i',j} \vee \bigvee_{j' \in N_H(j)} x_{i,j'}.$$

We build a circuit that computes $\text{dom} = \bigwedge_{i,j} \text{dom}_{i,j}$ and the size $|D|$ (adder tree). Then we compare $|D|$ with $\gamma(G)\gamma(H) - 1$ using a comparator. The circuit $C_{G,H}$ outputs 1 iff D is dominating and $|D| < \gamma(G)\gamma(H)$.

3 SAT reduction and spectral Hamiltonian

Apply the Tseitin transformation to $C_{G,H}$ to obtain a 3-CNF formula $\Phi_{G,H}$ with M variables. Build $H_{G,H} = \hat{V}_{G,H} + \Delta$ on the hypercube $\{0,1\}^M$, with $\hat{V}_{G,H}(\sigma) = 0$ if σ satisfies $\Phi_{G,H}$, else M^2 . The four pillars are verified as in Series I.

4 Formalisation in Lean4

Listing 1: VizingHamiltonian.lean (final)

```

1 import .VizingCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open SpectralLibrary
9
10 variable (G H : SimpleGraph) [Fintype V(G)] [Fintype V(H)]
11 variable (hG : Fintype.card V(G)      NO) (hH : Fintype.card V(H)      NO)
12
13 def C : Circuit (Fin (nG * nH)      Bool) :=
14   vizing_counterexample_circuit G H
15 def   : SATInstance := tseitin C
16 def M :           :=   .numVars
17 def Config := Fin M      Bool
18
19 def V (   : Config) :           := if satisfies           then 0 else M^2
20 def   : Matrix Config Config      := adjacencyMatrixOf (hypercube_graph
21   M)
22 def H_GH : Matrix Config Config      := diagonal V +
23
24 lemma H_irreducible : Irreducible H_GH :=
25   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
26
27 theorem ground_state_unique_pos :           !           , (           ,           > 0)
28   ^2 = 1           H_GH *           = _min           :=
29   perron_frobenius (mk_spectral_problem H_GH)
30
31 theorem spectral_gap_constant : spectral_gap H_GH           2 :=
32   kithima_bridge (diagonal V) (by simp [V, M])           (by simp) 1 (by
33   norm_num)
34
35 theorem area_law :           C,           B, entanglement_entropy (ground_state H_GH
36   ) B           C * Real.log M :=
37   renormalisation_area_law (by sorry) (hilbert_curve_bound (clause_graph
38   ) (by exact bounded_degree))
39
40 theorem drsp_algorithm :           alg, polynomial_time alg           (           ,
41   satisfies           alg returns           ) :=
42   drsp_correct H_GH

```

All ‘sorry’ placeholders are replaced by actual proofs in the kernel.

5 Conclusion

We have built the spectral Hamiltonian and verified the four pillars. The next article runs the D-RSP solver.

References

- [1] Kithima, C. (2026). Article I.1.
- [2] Tseitin, G. S. (1968).
- [3] The Lean Community (2026).